File Analysis

File Types, Contents and Signatures

Hypothesis and Some Magic

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Overview

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2. Content Identification
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File Analysis

Identification | Examination | Recovery

What is File Analysis?

- Determining the nature of files identified and extracted
- Important in any investigation
  - E.g., checking malware entry points, document validation, etc.
- Activities
  - Content Identification / Examination / Recovery
  - Metadata Identification / Examination / Recovery

File Analysis: Content Identification

- Identifying and verifying what a **file** really is
  - *A file maintained in computer-readable form.*

- This skill is needed so that:
  - You **don't rely** on file extension to assume what a file contains
  - You can **analyze** what cached files and general data are (usually no extension)

File Analysis: Content Identification

Magic Numbers

- Structures inherent to a specific file type
- Hexadecimal values found at specific offsets from the beginning of a file

- Can usually be used to identify a file
  - Examples:
    - JPEG images always begin with 0xFFD8FFE0 (yoya)
    - PDF documents always begin with 0x25504446 (%PDF)
    - Window Binaries always begin with 0x4D5A (MZ)

File Analysis: Content Examination

- **Needed skill** in order to understand how magic value tools operate and *to confirm / validate correct operation*

- Requires **reading hexadecimal dumps (hex dumping) of file content** *if done manually*

File Analysis: Content Recovery

- Depending on the case, *files are often mangled* and *purposefully made unreadable* by attackers **to avoid detection**
  - **Deleting and/or encrypting files** may trigger alerts
  - If *undeleted*, malware and other file types may be discovered and investigated, which may lead to the whole **attack operation being compromised**
  - **Modifying files** to affect its readability is often beneficial for attackers
    - Evade detection since modifying a file changes its signatures
    - Discourage analysts by making the file unreadable

File Analysis: Content Recovery

- *Finding indicators* about a **file's original format**, with the goal of recovering the said file's contents
- It involves **manually fixing the file** based on a *hypothesis*
  - Example:
    - A file starts with 0x5089474E, and is currently unreadable
      - A PNG file starts with 0x89504E47
      - In order to recover the file and its contents, we may have to swap the bits in all, or some parts of the file
- Recovering a file's contents highly depend on the skills of both the attacker and the analyst

File Analysis: Metadata Extraction

- Information stored within the file itself that provide extended information about file content:
  - Timestamps
  - Author information, etc.
- Available metadata largely depends on the type of file examined

File Analysis: Metadata Extraction

Image Metadata

- EXIF (Exchangeable Image File Format)
  - Embeds information about the device capturing the image
- IPTC (Information Interchange Model)
  - Embeds information about images used by newspapers and news agencies
- XMP (eXtensible Metadata Platform)
  - Open and extensible metadata scheme commonly used for images

File Analysis: Metadata Extraction

Audio Metadata

- Primary interest is usually audio content, but metadata can be a source of artifacts too

- Metadata is usually parsed with exiftool or mediainfo

File Analysis: Metadata Extraction

Video Metadata

- Metadata usually carries information regarding video/audio encoding and programs used to generate the video file

File Analysis: Metadata Extraction

Archive File Metadata

- May contain small amounts of metadata - usually in the form of user comments

- Depending on archive format, may retain system timestamps and data related to system of origin

File Analysis: Metadata Extraction

Document File Metadata

- Refers to file type containing text, images, and rendering Information
- Metadata usually carries authorship information / document revision histories / timestamps and system information about the system(s)
- Metadata is involved when authenticating a document as genuine or when checking for malware presence

File Analysis: Metadata Extraction

Points for Consideration

- Older and simpler file formats are less likely to carry metadata
- Usually, metadata can be altered without affecting the function of the content of the file
- Absence of metadata can be an artifact in itself